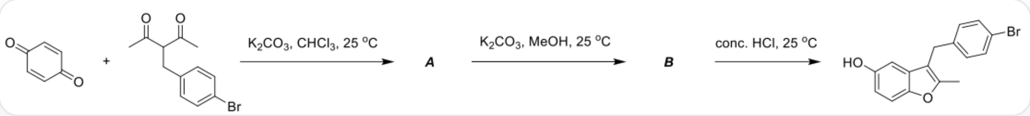
Question

In recent years, various one-pot multi-step reactions have been developed to construct aromatic heterocycles. Benzofuran, an important oxygen-containing heterocycle, is a key pharmacophoric scaffold for many bioactive compounds. The following reaction can construct 2,3-dialkyl-5-hydroxybenzofuran.



The reaction is represented by SMILES as, the first step O=C(C=C1)C=CC1=O.CC(C(C(C)=O)CC2=CC=C(Br)C=C2)=O>>[*], with conditions K_2CO_3, $CHCl_3$, 25 $^{\circ}C$, the second step of the reaction is the generation of **B** from **A**, with conditions K_2CO_3, $MeOH$, 25$^{\circ}C$, the third step of the reaction is [B]>>CC1=C(C2=CC(O)=CC=C2O1)CC3=CC=C(Br)C=C3, with conditions conc., HCl, 25$^{\circ}C$

Deduce the structures of intermediates **A** and **B**. Regarding **A** and **B**, the following statements are made:

1. **A** has three rings.

2. The generation of **A** involves a tricyclic intermediate.

3. **A** contains four oxygen atoms.

4. The generation of **B** from **A** involves an enolate intermediate.

5. The generation of the product from **B** involves a hemiketal intermediate.

6. The driving force of the entire reaction lies in aromatization.

Let a be the sum of the squares of the serial numbers of the correct statements, and b be the square of the sum of the incorrect statements. Find a/b, rounded to three significant figures.

A. All other options are incorrect

B. 0.285

C. 0.380

D. 0.490

E. 0.313

F. 0.953

G. 0.194

H. 0.610

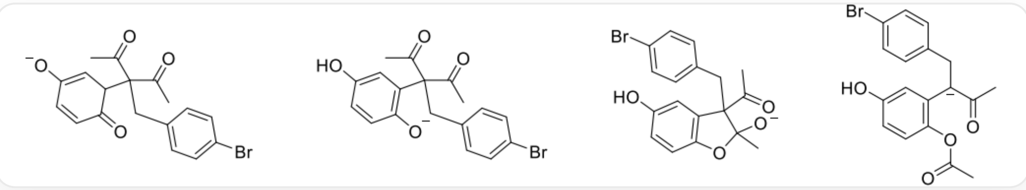
Answer

Correct Answer: D

Detailed Explanation

First, in chloroform, the common α-hydrogen of the diketone is removed by a base, forming an enolate anion. The carbon end attacks the β-position carbon of the carbonyl group on the benzoquinone, forming an enolate anion on the benzoquinone, followed by rearomatization to form a hydroquinone anion structure. Subsequently, the phenoxide anion attacks the carbonyl carbon in the β-diketone, forming a five-membered ring and a hemiketal anion structure, followed by ring-opening, forming an enolate anion, and **A** is obtained after workup.

The intermediates for the formation of **A** are shown below.



Intermediates for the formation of **A**. The first intermediate is represented by SMILES as CC(=O)C(CC1=CC=C(C=C1)Br)(C(=O)C)C2C=C(C=CC2=O)[O-]; The second intermediate is represented by SMILES as CC(=O)C(CC1=CC=C(C=C1)Br)(C(=O)C)C2=CC(=CC=C2[O-])O; The third intermediate is represented by SMILES as CC(=O)C1(CC2=CC=C(C=C2)Br)C3=CC(=CC=C3OC1(C)[O-])O; The fourth intermediate is represented by SMILES as CC([C-](C1=C(OC(C)=O)C=CC(O)=C1)CC2=CC=C(Br)C=C2)=O.

CHECKPOINT 1 PTS

The first intermediate for the formation of **A** is represented by SMILES as CC(=O)C(CC1=CC=C(C=C1)Br)(C(=O)C)C2C=C(C=CC2=O)[O-]

CHECKPOINT 1 PTS

[The second intermediate for the formation of **A** is represented by SMILES as CC(=O)C(CC1=CC=C(C=C1)Br)(C(=O)C)C2=CC(=CC=C2[O-])O

CHECKPOINT 1 PTS

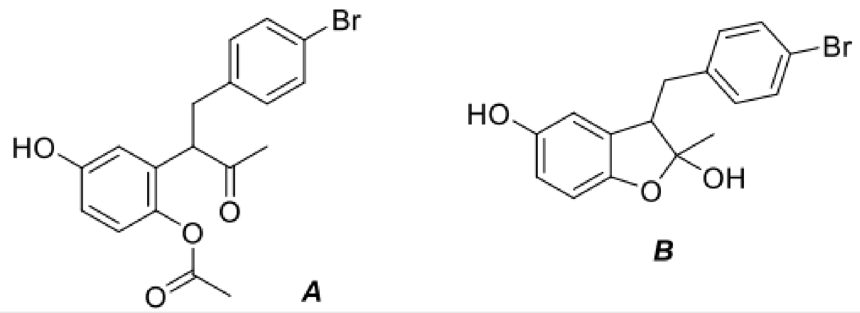
The third intermediate for the formation of **A** is represented by SMILES as CC(=O)C1(CC2=CC=C(C=C2)Br)C3=CC(=CC=C3OC1(C)[O-])O

CHECKPOINT 1 PTS

The fourth intermediate for the formation of **A** is represented by SMILES as CC([C-](C1=C(OC(C)=O)C=CC(O)=C1)CC2=CC=C(Br)C=C2)=O

In protic solvents, **A** loses an acetyl group to form a phenoxide anion, and then the phenoxide anion attacks the ketone carbon to generate a hemiketal anion, and hemiketal **B** is obtained after workup.

The structures of **A** and **B** are represented by SMILES as CC(C(CC1=CC=C(C=C1)Br)C2=C(C=CC(O)=C2)OC(C)=O)=O and CC1(O)C(CC2=CC=C(C=C2)Br)C3=C(C=CC(O)=C3)O1, respectively, as shown in the figure below.



The structures of **A** and **B** are represented by SMILES as CC(C(CC1=CC=C(C=C1)Br)C2=C(C=CC(O)=C2)OC(C)=O)=O and CC1(O)C(CC2=CC=C(C=C2)Br)C3=C(C=CC(O)=C3)O1

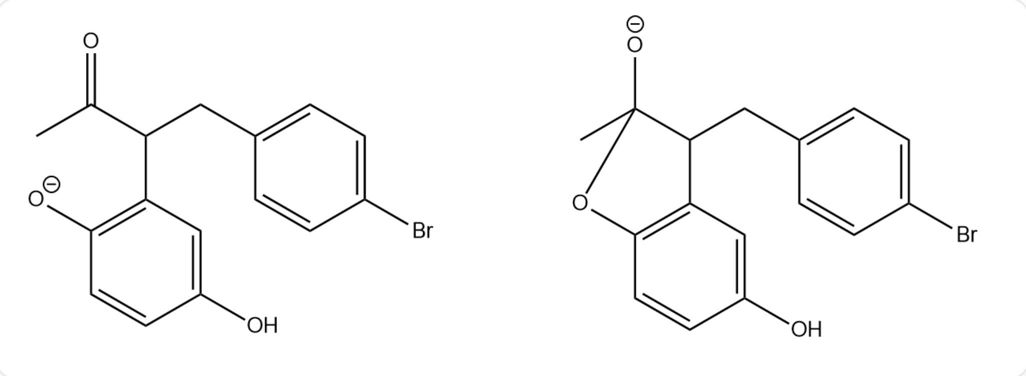
CHECKPOINT 2 PTS

The structure of **A** is represented by SMILES as CC(C(CC1=CC=C(C=C1)Br)C2=C(C=CC(O)=C2)OC(C)=O)=O

CHECKPOINT 2 PTS

The structure of **B** is represented by SMILES as CC1(O)C(CC2=CC=C(C=C2)Br)C3=C(C=CC(O)=C3)O1

The transformation from **A** to **B** involves two intermediates as shown in the figure below.



Two intermediates for the formation of **B** from **A**. The first intermediate is represented by SMILES as CC(C(CC1=CC=C(C=C1)Br)C2=C(C=CC(O)=C2)[O-])=O; The second intermediate is represented by SMILES as CC1([O-])C(CC2=CC=C(C=C2)Br)C3=C(C=CC(O)=C3)O1.

CHECKPOINT 1 PTS

The first intermediate for the formation of **B** from **A** is represented by SMILES as CC(C(CC1=CC=C(C=C1)Br)C2=C(C=CC(O)=C2)[O-])=O

CHECKPOINT 1 PTS

The second intermediate for the formation of **B** from **A** is represented by SMILES as CC1([O-])C(CC2=CC=C(C=C2)Br)C3=C(C=CC(O)=C3)O1

Subsequently, **B** undergoes dehydration under concentrated, acidified conditions to form a benzofuran product.

CHECKPOINT 1 PTS

Dehydration of **B** to generate the product

Now, determine whether each statement is correct or not. **A** has two rings, statement 1 is incorrect. The third intermediate for the formation of A has 3 rings, statement 2 is correct. **A** has 4 oxygen atoms, statement 3 is correct. The formation of **B** from **A** does not involve an enolate anion, statement 4 is incorrect. The formation of the product from **B** only involves an oxonium ion intermediate, statement 5 is incorrect. A benzene ring is formed during the formation of **A**, and a furan ring is formed during the formation of **B**. Aromatization is an important driving force for the reaction, statement 6 is correct.

The sum of the squares of the correct statement numbers is a, and the sum of the squares of the incorrect statement numbers is b. $a = 4 + 9 + 36 = 49$, $b = (1 + 4 + 5)^2 = 100$, $a/b = 0.490$, choose D.